Photometric Test Report

Reference Standards

ANSI C82.77-10-2014
IES LM-79-2019

Prepared For

Foshan MLS Technology Co., Ltd.

No. 1-4 Industrial Road (North), Chancheng District, Foshan, Guangdong, 528000, CN
Vincent Deng, vincent.deng@ledvance.cn

Test Laboratory:
UL-CCIC Company Limited
Test Laboratory Address:
No.2, Chengwan Road, Suzhou Industrial Park, Suzhou 215122, China
Catalog Number
BPANEL-50S54-UNV-D-8SC61-14-G-WH-H

Project Number
4791498147.3
Report Number
4791498147.3-8a

Test Date
2024-10-30~2024-11-05
Issue Date
2024-11-21
Revision Date
N/A

Prepared By	Approved By
Becky Fan	Roger Xu

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Test List

Sample Received Date: 2024-10-16

Test Item	Sample ID	Model Number	Tests Conducted By
Integrating Sphere Test	7712172	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Hom Dong
Goniophotometer Test	7712172	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Hom Dong
THD and PF Test	7712172	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Hom Dong

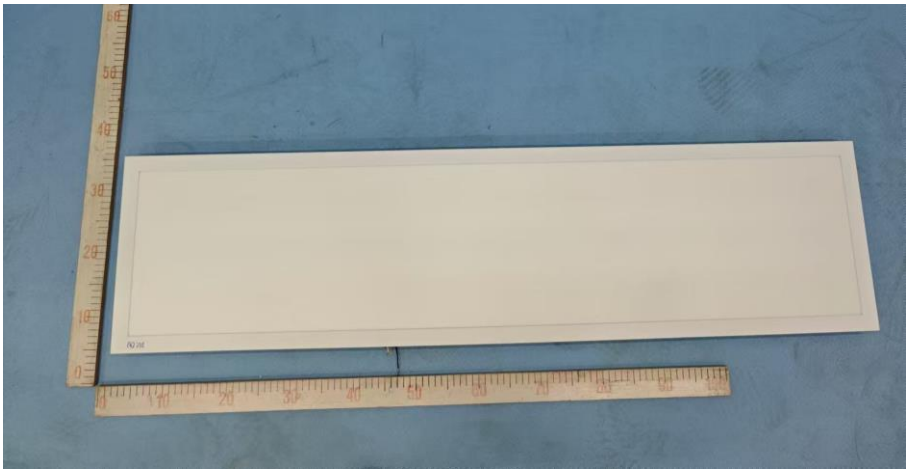
Remark (if any)

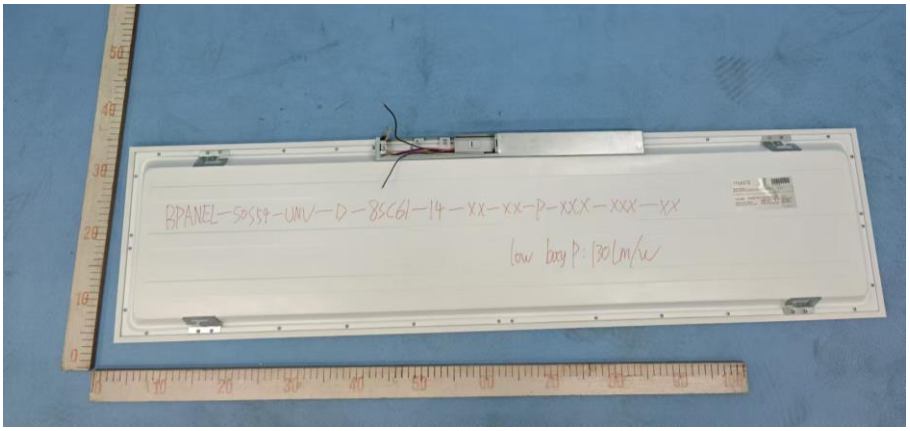
1. UL test equipment information is recorded on Meter Use in UL's Aurora database.

Product Description

Luminaire Description: Low Bay Luminaires (Commercial and Industrial)
Model Number: BPANEL-50S54-UNV-D-8SC61-14-G-WH-H
Electrical Ratings and CCT: 120-277Vac, 50/60 Hz, 50W max, 3000K/3500K/4000K/5000K/5700K/6500K tunable
Driver Model Number: BQ-50L6-1250-40-M56
LED Package: BXVN-30E-13H-9NV, BXVN-65E-13H-9NV, Bridgelux Inc
Family Model and Variation: BPANEL-50S54-UNV-D-8SC61-14-XX-YY-H-ZZZ-***-##, where "XX" represent mounting, can be G=Grid mounting, SF=Surface mounting, SP=Suspended mounting, DF=Drywall flange mounting. "YY" represent Finish color, can be WH=White or others. "ZZZ" represent Emergency, can be EM*= Emergency types(* can be blank or any letter), blank=No emergency. "***" represent Sensor receptacle, can be SR#= Sensor receptacle types(# can be blank or any number), blank= Without sensor receptacle. ## represent junction box, can be JB=with junction box, blank=without junction box.

Photos of Products Characteristics





Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Sample ID.	7712172
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

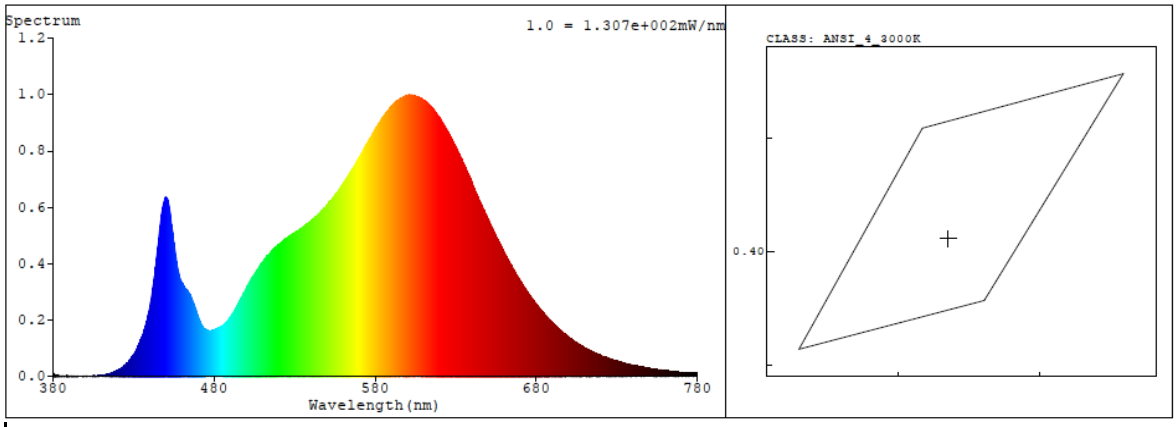
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.415	49.32	0.9913	Horizontal

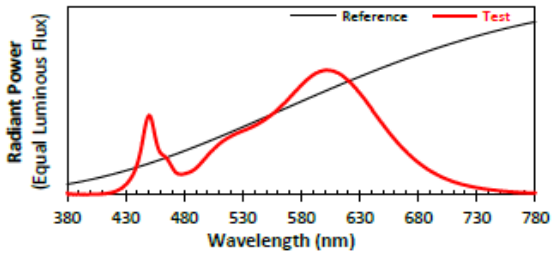
Test Results

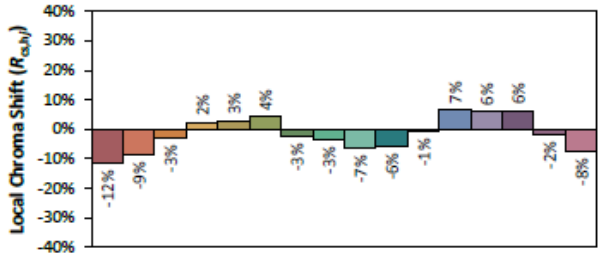
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3033	82	3	84	97	6400.6	129.8

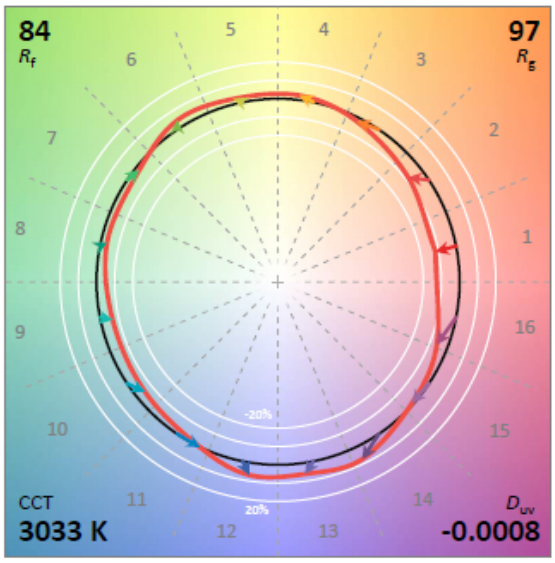
Spectrum

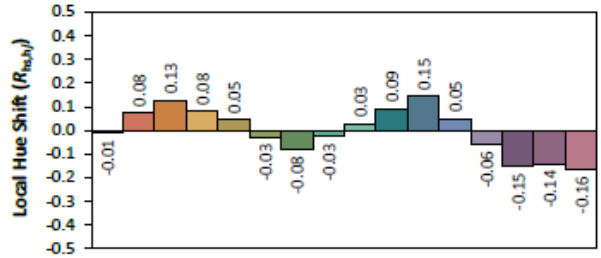


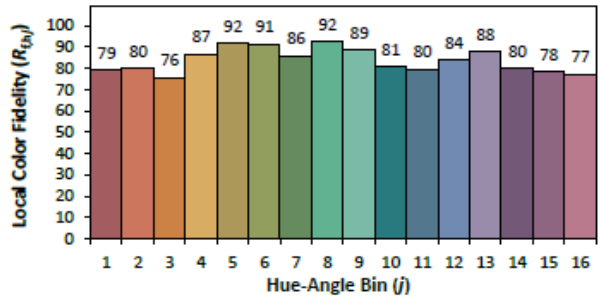
ANSI/IES TM-30-18 Color Rendition Report

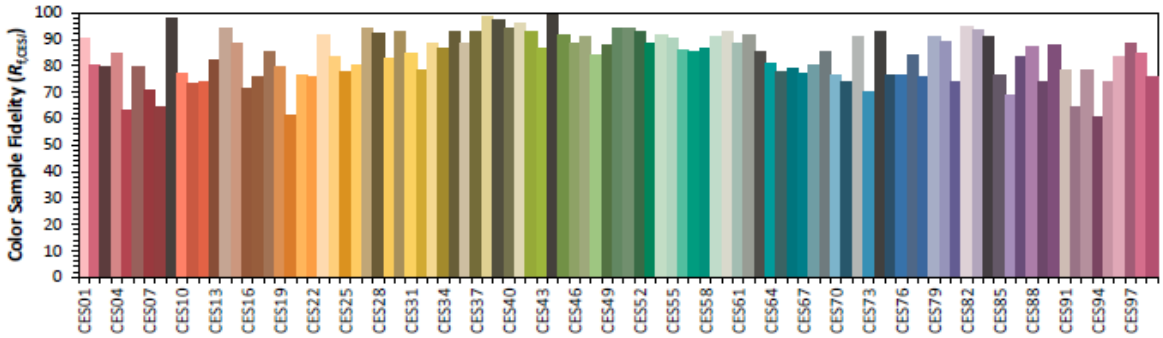












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4335
y	0.4010
u'	0.2497
v'	0.5197

CIE 13.3-1995 (CRI)	
R_a	82
R_9	3

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Sample ID.	7712172
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

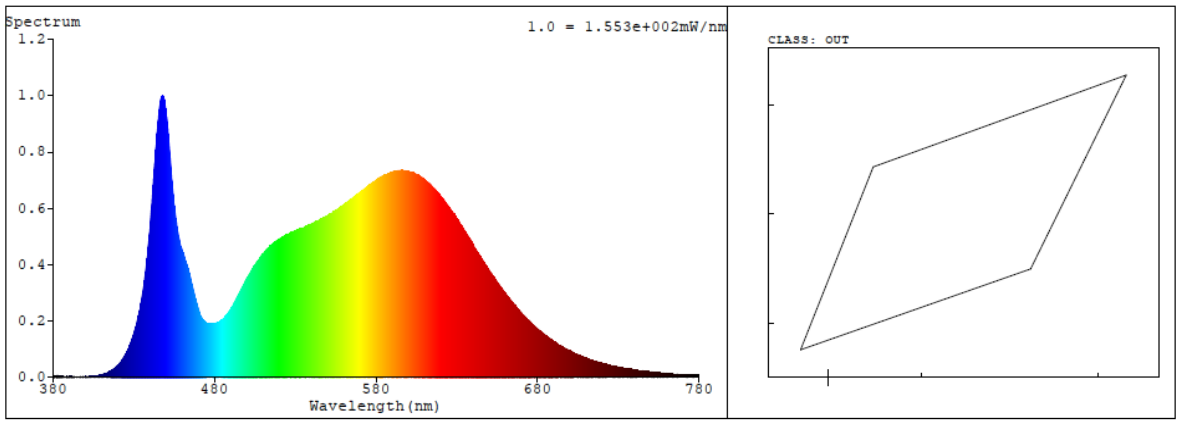
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.414	49.2	0.9913	Horizontal

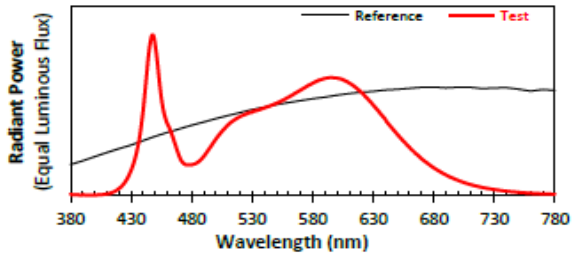
Test Results

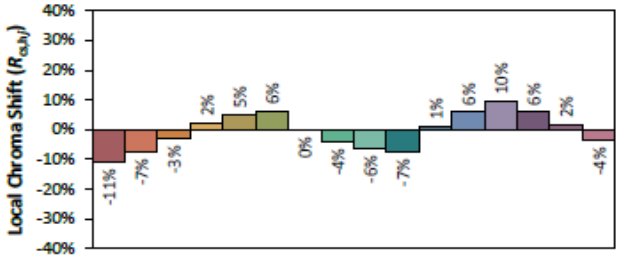
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
4070	84	17	84	99	6533.2	132.8

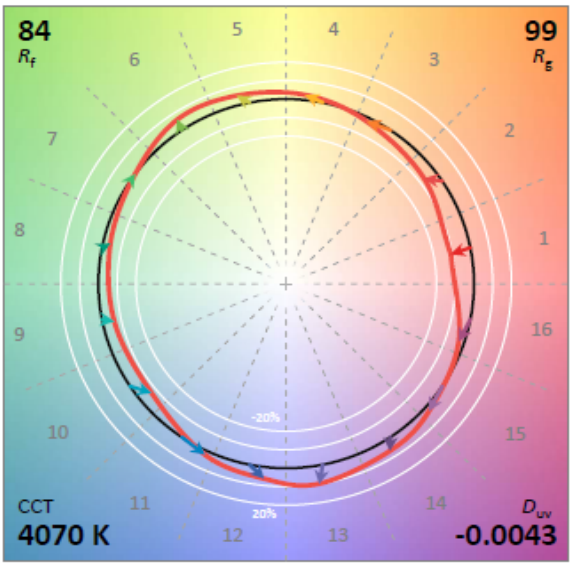
Spectrum

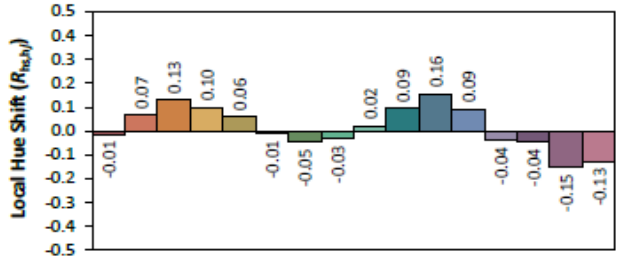


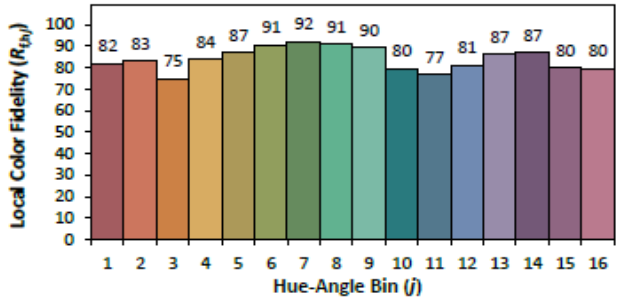
ANSI/IES TM-30-18 Color Rendition Report

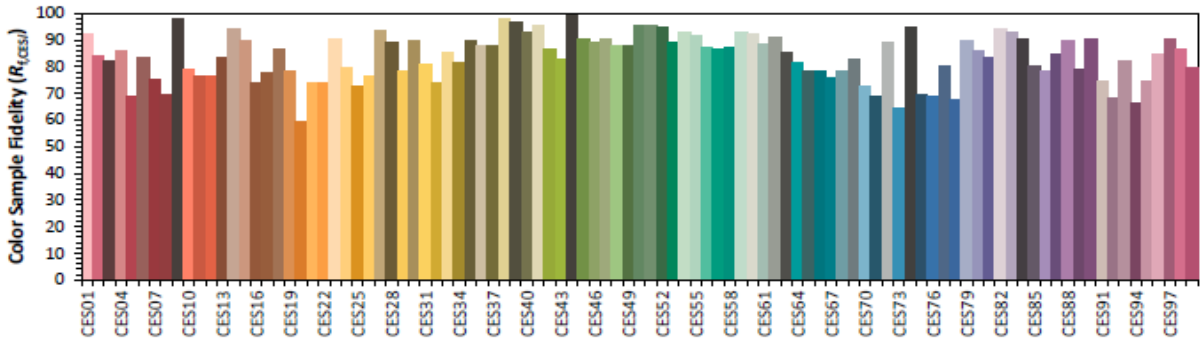












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3747
y	0.3642
u'	0.2263
v'	0.4951

CIE 13.3-1995 (CRI)	
R_a	84
R_9	17

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Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Sample ID.	7712172
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
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3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

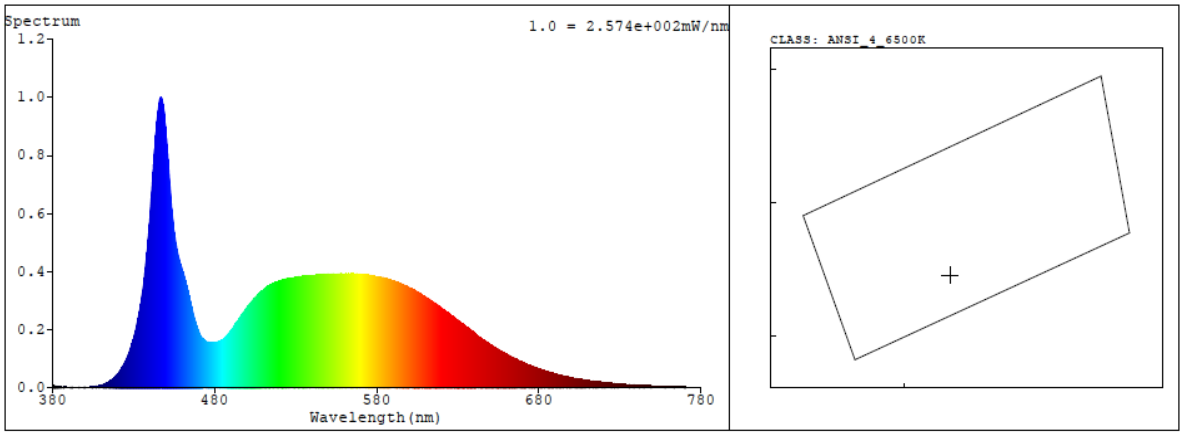
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.418	49.69	0.9915	Horizontal

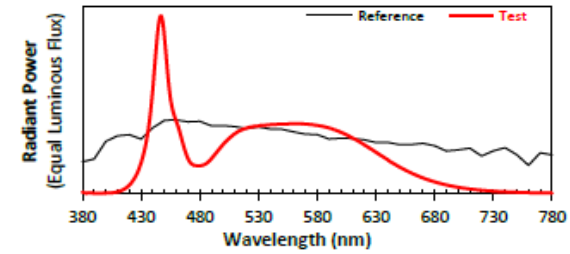
Test Results

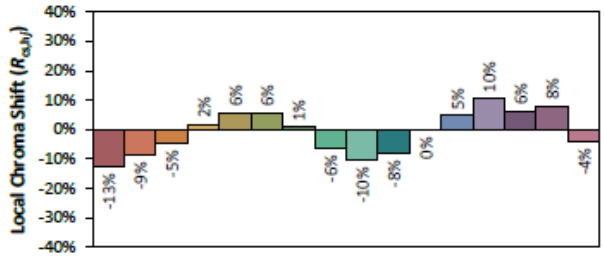
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
6599	81	7	81	97	6585.2	132.5

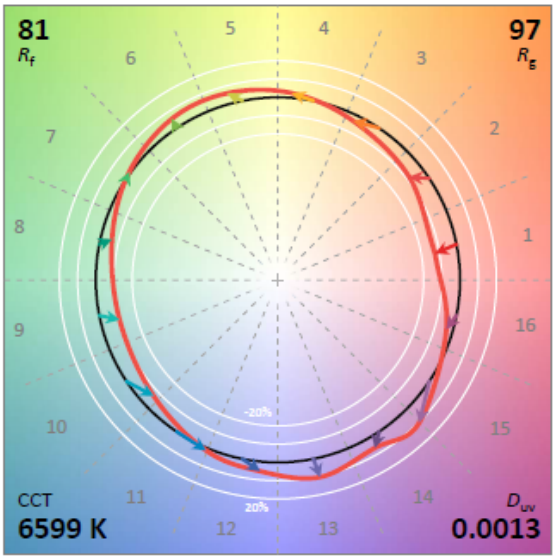
Spectrum

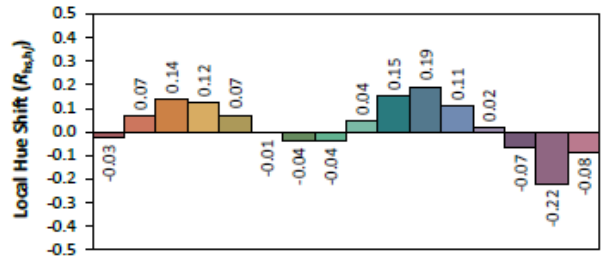


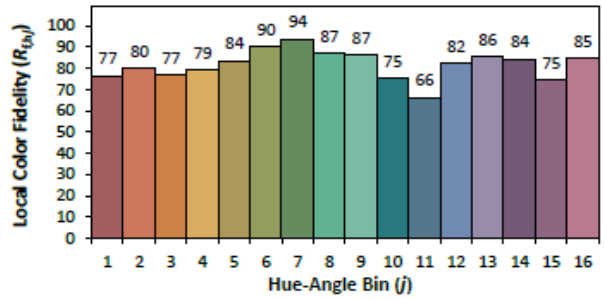
ANSI/IES TM-30-18 Color Rendition Report

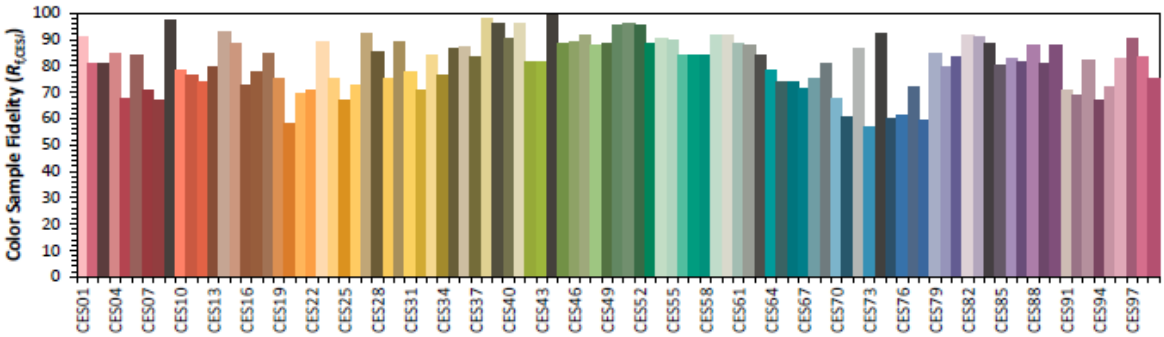












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3117
y	0.3243
u'	0.1989
v'	0.4656

CIE 13.3-1995 (CRI)

R_a	81
R_9	7

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Goniophotometer Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Sample ID.	7712172
Operate time (Min.)	65	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric parameters were measured using a type C goniophotometer and software.
3.The ambient temperature shall be maintained at 25 °C ± 0.9 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The reference standard lamp is power 400W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
4.The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, luminaire efficacy, zonallumen were calculated from the software taken at 1° vertical intervals and 22.5° horizontal intervals. Photometric distance was more than five times of the largest dimension of the test SSL product.

Goniophotometer Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.3	120.1	60	0.415	49.52	0.9928	Horizontal

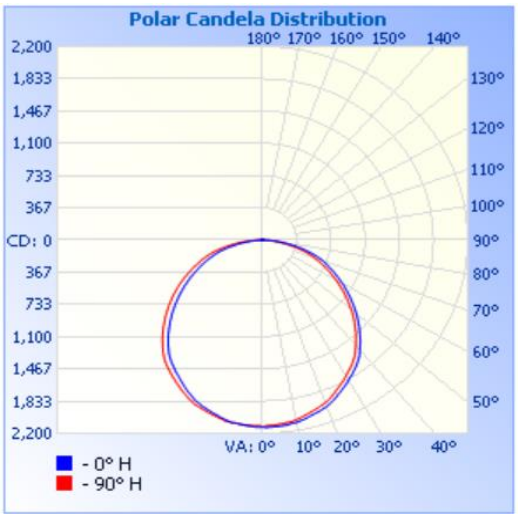
Test Results

Zonal Lumen Requirement (20°-50°)	Field Angle (10%)		Beam Angle (50%)		Flux (lm)	Luminous Efficacy (lm/W)
	Horizontal Spread	Vertical Spread	Horizontal Spread	Vertical Spread		
48.4%	165.6	165.1	114.5	113.3	6371.4	128.7

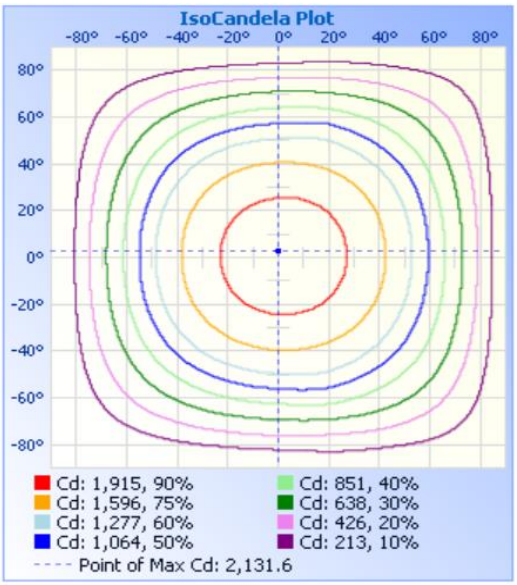
UGR
24

Goniophotometer Test (Cont'd)

Polar Candela Distribution



IsoCandela Plot



Goniophotometer Test (Cont'd)

Zonal Lumen Summary

Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	1,663.0	26.1%
0-40	2,741.9	43%
0-60	4,879.3	76.6%
60-90	1,473.4	23.1%
70-100	680.2	10.7%
90-120	9.1	0.1%
0-90	6,352.7	99.7%
90-180	18.7	0.3%
0-180	6,371.4	100%

Lumens Per Zone

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	201.3	3.2%	90-100	5.1	0.1%
10-20	578.5	9.1%	100-110	2.0	0%
20-30	883.2	13.9%	110-120	2.1	0%
30-40	1,079.0	16.9%	120-130	2.2	0%
40-50	1,123.0	17.6%	130-140	2.2	0%
50-60	1,014.3	15.9%	140-150	2.0	0%
60-70	798.3	12.5%	150-160	1.6	0%
70-80	511.6	8.0%	160-170	1.2	0%
80-90	163.5	2.6%	170-180	0.5	0%

Goniophotometer Test (Cont'd)

Intensity Data(cd)

	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124	2124
1	**2132**	2125	2118	2115	2113	2115	2121	2128	2130	2125	2119	2113	2110	2118	2124	2126	2127
2	2129	2126	2117	2112	2109	2115	2120	2127	2127	2126	2117	2114	2113	2117	2125	2127	2125
3	2127	2125	2118	2113	2112	2113	2118	2126	2122	2119	2117	2116	2113	2119	2125	2129	2130
4	2127	2122	2113	2111	2107	2107	2112	2119	2118	2121	2120	2114	2114	2120	2125	2124	2127
5	2125	2121	2107	2104	2100	2107	2111	2118	2115	2117	2116	2113	2113	2118	2126	2125	2125
6	2123	2114	2108	2100	2101	2099	2107	2112	2108	2116	2110	2114	2113	2117	2118	2126	2115
7	2120	2113	2105	2094	2092	2093	2104	2110	2106	2112	2110	2105	2105	2114	2120	2126	2117
8	2112	2109	2102	2097	2091	2089	2102	2103	2101	2109	2110	2109	2102	2111	2117	2121	2115
9	2107	2107	2099	2087	2085	2086	2095	2100	2102	2099	2103	2104	2106	2109	2119	2119	2109
10	2107	2097	2090	2084	2075	2079	2086	2092	2093	2092	2096	2095	2094	2103	2111	2105	2106
11	2097	2091	2084	2070	2066	2067	2072	2076	2081	2083	2090	2091	2096	2100	2106	2103	2094
12	2089	2079	2070	2062	2054	2056	2065	2071	2073	2078	2079	2079	2083	2088	2095	2091	2090
13	2081	2074	2060	2051	2044	2041	2048	2054	2059	2066	2069	2074	2077	2081	2090	2087	2080
14	2069	2060	2049	2041	2034	2037	2040	2045	2048	2058	2058	2066	2065	2073	2076	2079	2069
15	2061	2049	2039	2027	2021	2019	2023	2033	2037	2049	2050	2056	2061	2067	2072	2069	2060
16	2050	2042	2026	2021	2012	2010	2016	2024	2026	2043	2043	2044	2049	2053	2059	2066	2052
17	2044	2031	2019	2006	2003	1998	2005	2009	2015	2030	2032	2037	2044	2049	2053	2052	2042
18	2035	2019	2009	1997	1990	1989	1995	1999	2005	2019	2020	2028	2032	2040	2044	2046	2030
19	2023	2011	1998	1986	1979	1974	1985	1990	1996	2004	2015	2016	2025	2032	2039	2032	2024
20	2015	1999	1987	1976	1969	1965	1970	1976	1982	1988	2001	2007	2014	2020	2024	2019	2009
25	1930	1921	1907	1892	1879	1880	1884	1886	1898	1910	1923	1936	1945	1951	1952	1943	1929
30	1843	1828	1810	1796	1789	1782	1783	1786	1795	1828	1831	1852	1862	1869	1869	1865	1841
35	1750	1728	1711	1698	1688	1680	1686	1690	1702	1709	1742	1758	1772	1776	1780	1761	1746
40	1622	1610	1596	1575	1558	1553	1550	1556	1573	1591	1618	1644	1660	1664	1660	1639	1622
45	1472	1462	1443	1423	1403	1399	1395	1399	1418	1460	1466	1492	1515	1522	1514	1507	1472
50	1315	1303	1284	1262	1246	1234	1233	1234	1253	1317	1311	1337	1362	1369	1358	1369	1317
55	1152	1134	1117	1092	1077	1067	1065	1069	1085	1147	1150	1175	1199	1206	1196	1219	1151
60	990	968	950	924	913	899	896	903	920	983	984	1012	1036	1043	1033	1050	987
65	831	807	785	758	746	730	734	738	756	819	825	853	870	877	868	873	823
70	687	655	631	607	594	579	581	588	604	648	671	704	718	717	714	709	673
75	511	495	472	453	435	426	424	432	452	470	511	539	560	565	558	535	516
80	323	308	289	273	252	246	242	251	271	307	323	347	369	376	364	357	328
85	163	146	128	110	94	88	87	96	113	156	157	178	194	199	192	201	163
90	32	20	9	4	3	2	3	3	4	24	25	38	50	55	51	66	29
95	2	3	3	2	1	2	2	2	2	3	3	2	2	2	3	3	2
100	3	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
105	2	2	2	2	2	2	2	2	2	1	2	2	1	2	2	2	2
110	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
115	2	2	3	2	2	2	2	2	2	2	2	2	2	2	2	2	2
120	2	3	3	2	2	2	3	2	2	2	2	2	2	2	2	2	2
125	3	3	3	2	3	3	3	3	3	2	2	2	2	3	2	2	3
130	3	2	2	3	3	3	2	3	3	2	2	2	2	2	3	3	3
135	2	3	3	3	3	3	3	3	2	3	3	3	3	2	3	3	2
140	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
145	4	4	4	3	4	4	4	3	4	3	3	3	3	4	3	3	4
150	3	3	4	3	3	3	3	4	3	3	4	3	3	3	4	4	3
155	4	3	3	3	4	4	4	3	4	4	3	4	4	4	3	3	4
160	4	3	4	4	4	3	4	3	4	4	3	4	4	4	4	3	4
165	5	4	4	4	5	4	4	4	5	4	4	4	4	4	4	4	4
170	5	5	5	4	5	4	5	5	5	6	5	5	5	4	5	4	5
175	6	6	6	5	5	5	5	5	6	6	5	5	5	4	5	5	6
180	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

Corrected UGR values generated per CIE 190-2010

UGR Table - Corrected

Reflectances											
Ceiling Cavity		70	70	50	50	30	70	70	50	50	30
Walls		50	30	50	30	30	50	30	50	30	30
Floor Cavity		20	20	20	20	20	20	20	20	20	20
Room Size		UGR Viewed Crosswise					UGR Viewed Endwise				
X=2H	Y=2H	18.9	20.6	19.3	20.9	21.2	18.3	20.0	18.7	20.3	20.6
	3H	21.0	22.5	21.3	22.8	23.1	20.2	21.7	20.6	22.1	22.4
	4H	21.8	23.2	22.2	23.6	24.0	21.0	22.4	21.4	22.8	23.1
	6H	22.5	23.8	22.9	24.2	24.6	21.5	22.9	21.9	23.2	23.6
	8H	22.7	24.0	23.1	24.4	24.8	21.7	23.0	22.1	23.4	23.8
	12H	22.9	24.1	23.3	24.5	24.9	21.8	23.0	22.2	23.4	23.8
4H	2H	19.6	21.0	20.0	21.4	21.7	19.0	20.4	19.4	20.8	21.1
	3H	21.9	23.1	22.3	23.5	23.9	21.1	22.3	21.5	22.7	23.1
	4H	22.9	24.0	23.3	24.4	24.8	22.0	23.1	22.5	23.6	24.0
	6H	23.7	24.7	24.1	25.1	25.6	22.7	23.7	23.2	24.1	24.6
	8H	24.0	24.9	24.4	25.3	25.8	22.9	23.8	23.4	24.3	24.7
	12H	24.3	25.1	24.7	25.5	26.0	23.1	23.9	23.5	24.3	24.8
8H	4H	23.3	24.2	23.7	24.6	25.1	22.4	23.3	22.9	23.8	24.2
	6H	24.2	25.0	24.7	25.5	26.0	23.2	24.0	23.7	24.5	24.9
	8H	24.6	25.3	25.2	25.8	26.3	23.5	24.2	24.0	24.7	25.2
	12H	25.0	25.6	25.5	26.1	26.7	23.7	24.3	24.2	24.8	25.3
12H	4H	23.3	24.2	23.8	24.6	25.1	22.5	23.3	23.0	23.8	24.2
	6H	24.3	25.0	24.9	25.5	26.0	23.3	24.0	23.8	24.5	25.0
	8H	24.8	25.4	25.3	25.9	26.5	23.6	24.2	24.1	24.7	25.3

Maximum UGR = 26.7

THD and PF Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-H	Sample ID.	7712172

Test Method

1. The samples were tested according to the ANSI C82.77-10-2014.
2. The ambient temperature condition was maintained at 25 °C ± 1 °C. The sample measurement was made using a digital power meter and power supply. The sample was operated at rated voltage and stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

Test Results

CCT Range	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Current THD	Orientation
0%-3000K	120	60	0.415	49.5	0.9928	10.39%	Horizontal
	277	60	0.181	48.3	0.9654	8.73%	Horizontal
20%-3500K	120	60	0.414	49.3	0.9927	10.49%	Horizontal
	277	60	0.180	48.1	0.965	8.74%	Horizontal
40%-4000K	120	60	0.414	49.4	0.9927	10.42%	Horizontal
	277	60	0.180	48.2	0.9651	8.73%	Horizontal
60%-5000K	120	60	0.415	49.5	0.9927	10.41%	Horizontal
	277	60	0.181	48.3	0.9653	8.73%	Horizontal
80%-5700K	120	60	0.416	49.5	0.9928	10.41%	Horizontal
	277	60	0.181	48.4	0.9654	8.72%	Horizontal
100%-6500K	120	60	0.419	49.9	0.9929	10.35%	Horizontal
	277	60	0.182	48.7	0.9661	8.54%	Horizontal

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